

Polyhedra=finitely generated sets

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A polyhedron is the solution set to a finite number of linear inequalities, and a finitely generated set is a set of the form $\text{conv}(X) + \text{cone}(Y)$, where $X, Y \subset \mathbb{R}^n$ are finite sets. In this note we show that polyhedra and finitely generated sets are exactly the same objects.

Finitely generated cones are the same objects as polyhedral cones (proved in note 1.(F)):

Fact 1 (*Finitely generated cones=polyhedral cones*)

Given a matrix $A \in \mathbb{R}^{m \times n}$, there exists $E \subset \mathbb{R}^n$ finite such that $\{x \in \mathbb{R}^n : Ax \leq 0_m\} = \text{cone}(E)$, and conversely, given $E \subset \mathbb{R}^n$ finite, there exists a matrix $A \in \mathbb{R}^{m \times n}$ such that $\{x \in \mathbb{R}^n : Ax \leq 0_m\} = \text{cone}(E)$ holds. ■

Here we generalize Fact 1 to polyhedra and finitely generated sets:

Theorem 1 (*Finitely generated sets=polyhedra*)

Given a non-empty polyhedron $\{x \in \mathbb{R}^n : Ax \leq b\}$, where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$, there exists two non-empty finite sets $V, E \subset \mathbb{R}^n$ such that:

$$\{x \in \mathbb{R}^n : Ax \leq b\} = \text{conv}(V) + \text{cone}(E). \quad (1)$$

Conversely, given non-empty finite sets $V, E \subset \mathbb{R}^n$, there exists $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$ such that (1) is satisfied. ■

1 Polyhedra are finitely generated sets

First let $\{x \in \mathbb{R}^n : Ax \leq b\}$ be a non-empty polyhedron, where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. We will identify non-empty and finite sets $V, E \subset \mathbb{R}^n$ such that (1) is satisfied. This will prove one direction of Theorem 1. The idea is to use the so-called *cone extension* of $\{x \in \mathbb{R}^n : Ax \leq b\}$:

$$C_p(A, b) := \left\{ \begin{pmatrix} x \\ \lambda \end{pmatrix} \in \mathbb{R}^{n+1} : Ax - \lambda b \leq 0_m \text{ and } \lambda \geq 0 \right\}.$$

obtained from $\{x \in \mathbb{R}^n : Ax \leq b\}$ by parameterizing the right-hand-sides with a non-negative variable, and then use the result for polyhedral cones (Fact 1). The cone extension $C_p(A, b)$ is connected to $\{x \in \mathbb{R}^n : Ax \leq b\}$ by the equivalence $Ax \leq b \iff \begin{pmatrix} x \\ 1 \end{pmatrix} \in C_p(A, b)$.

Since $C_p(A, b)$ is a polyhedral cone, Fact 1 shows $C_p(A, b)$ is finitely generated, *i.e.*, there exists vectors $\{(\frac{x^i}{\lambda_i}) : i \in I\} \subset C_p(A, b)$ such that:

$$C_p(A, b) = \text{cone}\left(\left\{\begin{pmatrix} x^i \\ \lambda_i \end{pmatrix} : i \in I\right\}\right),$$

where I is a finite index set. Since $C_p(A, b)$ is a polyhedral cone, the generators $\{(\frac{x^i}{\lambda_i}) : i \in I\}$ of $C_p(A, b)$ can be scaled with positive multipliers. Hence we can assume $\lambda_i \in \{0, 1\}$ for all $i \in I$.

Define $V := \{x^i : \lambda_i = 1\}$ and $E := \{x^i : \lambda_i = 0\} \cup \{0_n\}$. The set V is not empty. Indeed, if one had $V = \emptyset$, then $C_p(A, b) = \text{cone}(\{(\frac{x^i}{0}) : i \in I\})$, which implies $\lambda = 0$ for all $(\frac{y}{\lambda}) \in C_p(A, b)$. However, since $(\frac{y}{1}) \in C_p(A, b)$ for all $y \in \{x \in \mathbb{R}^n : Ax \leq b\}$, this contradicts that $\{x \in \mathbb{R}^n : Ax \leq b\}$ is non-empty.

To finish we argue $\{x \in \mathbb{R}^n : Ax \leq b\} = \text{conv}(V) + \text{cone}(E)$:

$$\begin{aligned} & y \in \{x \in \mathbb{R}^n : Ax \leq b\} \\ \iff & \begin{pmatrix} y \\ 1 \end{pmatrix} \in C_p(A, b) \\ \iff & \begin{pmatrix} y \\ 1 \end{pmatrix} = \sum_{v \in V} \alpha_v \cdot \begin{pmatrix} v \\ 1 \end{pmatrix} + \sum_{e \in E} \beta_e \cdot \begin{pmatrix} e \\ 0 \end{pmatrix} \quad \text{for some } \alpha \geq 0_{|V|} \text{ and } \beta \geq 0_{|E|} \\ \iff & y \in \text{conv}(V) + \text{cone}(E). \end{aligned}$$

2 Finitely generated sets are polyhedra

Now let $V, E \subset \mathbb{R}^n$ be finite and non-empty sets. We will find $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$ such that (1) is satisfied. This will prove the remaining direction of Theorem 1. For this direction of the proof, we use the cone extension of $\text{conv}(V) + \text{cone}(E)$, which is the following finitely generated cone:

$$C_f(V, E) = \text{cone}\left(\left\{\begin{pmatrix} v \\ 1 \end{pmatrix} : v \in V\right\}\right) + \text{cone}\left(\left\{\begin{pmatrix} e \\ 0 \end{pmatrix} : e \in E\right\}\right).$$

The finitely generated cone $C_f(V, E)$ is connected to $\text{conv}(V) + \text{cone}(E)$ by the equivalence $x \in \text{conv}(V) + \text{cone}(E) \iff (\frac{x}{1}) \in C_f(V, E)$. Since $C_f(V, E)$ is finitely generated, Fact 1 applies. Hence there exists $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$ so that $C_f(V, E) = \{(\frac{x}{\lambda}) \in \mathbb{R}^{n+1} : [A, -b] \cdot (\frac{x}{\lambda}) \leq 0_m\} = C_p(A, b)$. We now have:

$$Ax \leq b \iff \begin{pmatrix} x \\ 1 \end{pmatrix} \in C_p(A, b) = C_f(V, E) \iff x \in \text{conv}(V) + \text{cone}(E),$$

which shows $\{x \in \mathbb{R}^n : Ax \leq b\} = \text{conv}(V) + \text{cone}(E)$.